

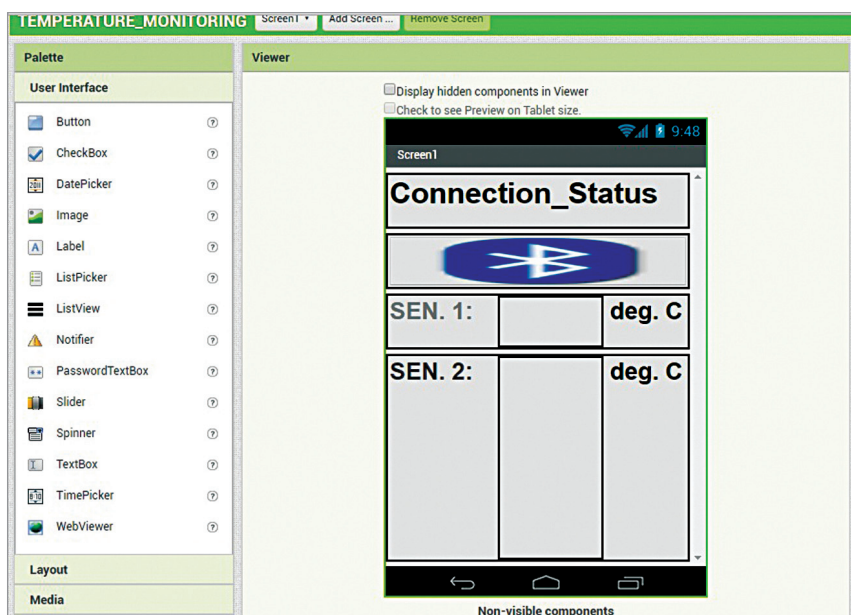


Fig. 4: Screenshot of the Designer window in MIT App Inventor

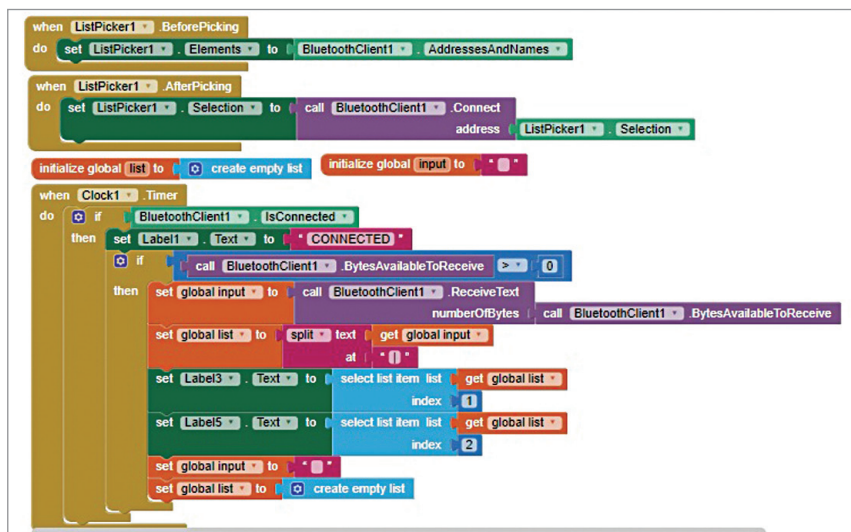


Fig. 5: Screenshot of the Blocks section in MIT App Inventor

AVR ATmega328P microcontroller-based development board with six analogue input pins and 14 digital I/O pins. The microcontroller has 32kB ISP flash memory, 2kB RAM, and 1kB EEPROM. The board provides capability of serial communication via UART, SPI, and I2C. The microcontroller can operate at 16MHz clock frequency. In this project, the Arduino serves two purposes: (a) converts the output voltage of the LM35s into a corresponding temperature value in degree centigrade, and (b) transmits the corresponding values to the Bluetooth module via Tx pin of the Arduino.

HC-05 Bluetooth module. It is an

easy-to-use Bluetooth SPP (Serial Port Protocol) module, which is designed for wireless serial connection setup. HC-05 can be used in a master or slave configuration, making it a great solution for wireless communication. It uses CSR Bluecore 04-External single-chip Bluetooth system with CMOS technology and AFH (adaptive frequency hopping) feature.

EFY Note
The source code
of this project
is available
for free download
at source.efymag.com

Android smart phone. The monitoring app (temp_multi_sensor.apk) is based on Android and has been designed using MIT App Inventor platform. The MIT App Inventor is an intuitive, visual programming environment that allows the user to build fully functional apps for smartphones and tablet PCs. When our designed app runs on the smartphone, the data sent by Bluetooth is received and displayed on the smartphone. The project has been tested on Redmi 3S and Realme 5 Pro smartphones.

Software

For execution of the project, we need following software programs:

1. The Arduino source code (temp_multi_sensor.ino), which should be uploaded on the Arduino Uno. The function Serial.println () helps in transmitting the temperature value from the Tx pin of the Arduino to Bluetooth.

2. The Android app (temp_multi_sensor.apk), which must be installed on the smartphone. After successful installation, turn on Bluetooth of the smartphone.

After successful pairing of HC-05 Bluetooth with Bluetooth of the smartphone, open the app and press the Bluetooth logo. From the list of paired devices shown in the app, choose the HC-05 device. After successful connection, 'Connected' message will be displayed on the app in the smartphone. The Android application will now display both the temperature data (SENS1 and SENS2).

A screenshot of the temperature sensed by the sensors (SENS1 and SENS2, as shown in Fig. 2) observed on the smartphone is shown in Fig. 3. Screenshots of the Designer and Blocks sections in MIT App Inventor are shown in Figs. 4 and 5, respectively. **EFY**



Aniket Subham has a keen interest in designing microcontroller based embedded systems



Shibendu Mahata is M.Tech (gold medallist) in Instrumentation and Electronics Engineering from Jadavpur University. His several papers in the domain of signal processing have been published in reputed journals and read at international conferences